

Classification of white blood cells using deep features obtained from Convolutional Neural Network models based on the combination of feature selection methods

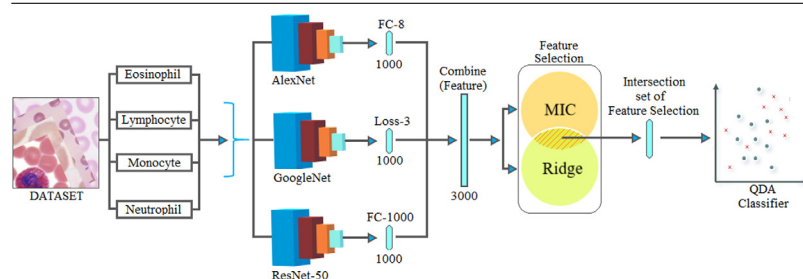
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GRAPHICAL ABSTRACT



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ABSTRACT

White blood cells are cells in the blood and lymph tissue produced by the bone marrow in the human body. White blood cells are an important part of the immune system. The most important task of these cells is to protect the human body against foreign invaders and infectious diseases. When the number of white blood cells in the blood is not enough for the human body, it can cause leukopenia. As a result of this situation, the resistance of the human body against infections and diseases decreases. In this respect, determining the number of these cells in the human body is a specialist task. Detection and treatment of this symptom is a labor-intensive process carried out by specialist doctors and radiologists. Image processing techniques have recently been widely used in biomedical systems for the diagnosis of various diseases. In this study, it is aimed to use image processing techniques to improve the classification performance of deep learning models in white blood cells classification. To perform the classification process more efficiently, the Maximal Information Coefficient and Ridge feature selection methods were used in conjunction with the Convolutional Neural Network models. The Maximal Information Coefficient and Ridge feature selection methods extracted the most relevant features. Afterward, the classification process was realized by using this feature set. In this study, AlexNet, GoogLeNet, and ResNet-50 were used as feature extractor and quadratic discriminant analysis was used as a classifier. As a result, the overall success rate was obtained as 97.95% in the classification of white blood cells. The experimental results showed that the use of the convolutional neural network models with feature selection methods contributed to improving the classification success of white blood cell types.

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1. Introduction

White blood cell (WBC) is a type of blood contained in the human body [1]. This type of blood cell plays an important role

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in the immune system of the human body. In other words, it fights against bacteria, viruses, and germs [2]. An adult human has about 4000 to 10,000 WBC. Symptoms of WBC disorder in the human body; chronic infections, rapid weight loss, weakness, fatigue may occur or trigger different diseases [2,3].

Early diagnosis is vital in such cases. The most common diagnostic method used in clinical examination of these cases is to take microscopic images of the patients and examine these images by specialist doctors [4,5]. WBC is generally divided into four classes namely Eosinophil, Lymphocyte, Monocyte, and Neutrophil as shown in Fig. 1. Detection of WBC types can be performed by microscopic imaging. This situation, the procedure to be followed in the type and count of a WBC depends on the experience and knowledge of the doctor [6]. This study aims to perform fast and accurate detection of WBC types with the proposed approach by using microscopic images. Therefore, the selection of features is important. It is aimed to select more productive features by using feature selection methods over feature sets obtained from deep learning models, and this situation is supported by an intersection approach in sets. In this way, the correct and fast classification of WBC according to its types will be realized with the proposed approach [7,8].

The type and count of patient WBCs play an important role in the diagnosis and management of many diseases. Convolutional Neural Network (CNN) based methods should be used as an effective solution for WBC classification [9–11]. Although CNN-based methods ensure high classification accuracy using the end-to-end learning schema, the achievement of such systems can be improved using deep features extracted in the different levels in CNN architecture. For this reason, the combination of feature selection methods can increase the WBC classification accuracy through more discriminant features in a lower-dimensional space. The combination of feature selection methods applied to different CNNs can increase the accuracy in WBC classification. In this paper, we propose and evaluate the combination of feature selection methods. Thus, the evaluation of single features vs. simple feature combination; evaluation of feature selection methods; by comparing the feature selection combination with the simple feature combination, it is purposed to achieve specific objectives such as estimating the best number of selected features.

The rest of this paper is organized as follows. A literature review related to WBC studies is given in Section 2. The dataset and methods are presented in Section 3. The proposed method is explained in Section 4. The experimental results are reported in Section 5. In Sections 6 and 7, discussion and conclusion remarks are given, respectively.

2. Literature review

In this section, a summary of the literature on the classification of WBC is presented considering the dataset, methods, and performance metrics.

K. Abbas et al. [12] separated WBC data into classes and subclasses. WBC data consisted of three main classes and seven subclasses. In the classification steps, they used shape recognition, image rotation techniques, pattern recognition methods, segmentation, feature extract, and deep learning models together. For the segmentation process, they used thresholding techniques, edge, and boundary detection and Geometric Active Contour (GACs) [13] and level set method. They used bi-spectral invariant features and L -moments invariant features for feature extraction. In the classification step, they used Support Vector Machines (SVM) [14] and Decision Tree (DT) [15]. The overall accuracy rate obtained in their studies was 97.23%. J. Zhao et al. [16] presented an automatic detection and classification method for WBC data. Microscope images were examined and WBC types

were determined by using a connection between red, green, blue (RGB), and intermediate colors of these colors. For classification of some types of WBC (basophil and eosinophil), granularity feature (pairwise rotation invariant co-occurrence local binary pattern) was used together with the SVM classifier. In the next step, they used the CNN model to produce features. Random Forest (RF) [17] was used to classify other WBC types (lymphocytes, monocytes, and neutrophils). The classification accuracy was limited to 92.6%. M. Habibzadeh et al. [18] performed the classification of WBC data in three steps. In the first step, they proposed three consecutive pre-processing calculations; bounding box distortion (crop) and image flipping mirroring. In the second step, white blood cell recognition was performed with a hierarchy topological feature extraction using Inception and ResNet architectures. And the final step was the classification process. They used the fine-tuned CNN model. As a result, they achieved 99.84% classification accuracy. F.I. Kurniadi et al. [19] classified the WBC data with VGG-16 model. They used the VGG-16 model together with the Local Binary Pattern (LBP) [20] for feature extraction. They used XG-Boost method [21] with the k -Nearest Neighbor (k NN) [22] in the classification process. The success rate obtained using the test data was 92.93%. They also performed a reclassification of the dataset using the augmentation method. The success rate obtained with augmented data was 90.16%. Q. Huang et al. [23] used hyperspectral images as the type of image in the classification of WBC dataset. They used the Gabor Wavelet method together with CNN in their study. For each convolution layer, they used multi-scale and orientation Gabor operators. In the next step, they combined the features of Gabor Wavelets [24] and performed the classification process by the SVM method. The success rate of classification in their studies was 97.65%. M.I. Razzak et al. [25] performed segmentation and classification using WBC dataset. The model in the classification step used Extreme Machine Learning (ELM) [26] method with CNN. The segmented cell images were masked before the feature extraction stage. The features were then extracted from each segmented cell. They achieved a success rate of 95.10%.

Considering the related studies, some of the works have realized classification task using handcrafted features. This is a labor-intensive process. As for CNN based approaches, pretrained CNN models have been adopted as a useful tool to separate different WBC types. Transfer learning techniques have been utilized to adopt CNNs to this new task. In this manner, fully automated systems have been introduced. In this study, we focus on the deep features extracted from different levels in the CNN architecture. The most relevant features in the three different pretrained CNN models are achieved by using the combination of feature selection methods. As a result, a robust and fully automated model with high sensitivity has been revealed for WBC classification.

3. Materials and methods

3.1. WBC dataset

The open-access dataset used in this study consists of four categories: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. The WBC dataset consists of 3,120 eosinophils, 3,102 lymphocytes, 3,091 monocytes, and 3,122 neutrophil microscopic images. So, the dataset ensures a totally of 12,435 images. Each image makes up of a depth of 24 bits and a resolution of 320×240 pixels. The file extension of the images is JPEG format. Each sample in the dataset was examined and labeled by experienced specialists [27].

The number of data in each class was kept equal for efficient use in this study [28]. Totally 12,364 images, 3,091 images from each class, were used in the experiments. In classes with images over 3,091, image selection was carried out randomly. In addition, the dataset was divided into two parts as training and test sets with 70% and 30% rates, respectively.

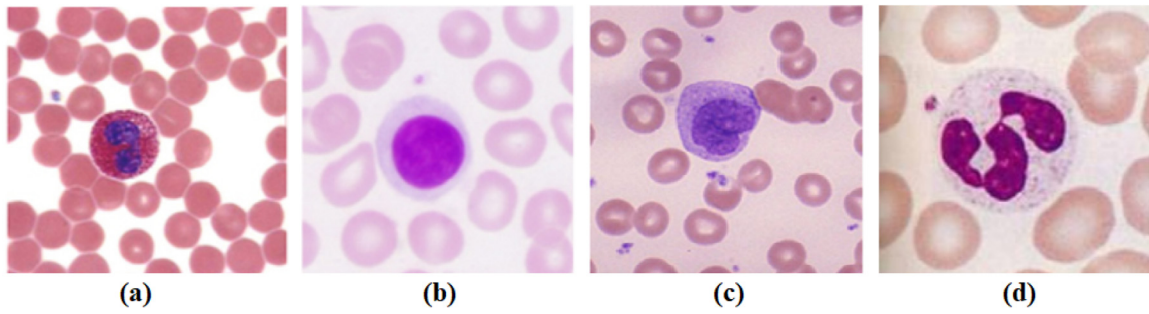


Fig. 1. Types of WBC; (a) Eosinophil, (b) Lymphocyte, (c) Monocyte and (d) Neutrophil.

3.2. CNN models

The common point of these three architectures (AlexNet, GoogLeNet, ResNet-50) is that they have fully connected layers providing 1000 extracted feature maps [29]. In this study, the features to be used in the classification phase were obtained from the FC-8 layer of the AlexNet model, the Loss3-classifier layer of the GoogLeNet model, and the FC-1000 layer of the ResNet-50 model. A total of 3000 features were obtained, 1000 features from each layer. The schematic representation of the CNN architectures is shown in Fig. 2.

AlexNet is one of the most important CNN architectures. AlexNet became well-known after winning the 2012 ImageNet competition with a top-five test error rate of 15.3%. This architecture consists of convolution, pooling, and FC layers. The input size of this model is 227×227 pixels. The size of convolutional filters is typically 3×3 or 5×5 pixels [8,24]. With these filters, more efficient feature maps are extracted. The most important feature of extracted feature maps is that they have learned features [30]. The pooling layer is used to reduce the size and cost of the image by interacting with its features [31].

GoogLeNet was announced in the 2014 ImageNet competition. The network structure of GoogLeNet is complex. In addition to convolution and pooling layers, the GoogLeNet architecture includes a stack set consisting of a combination of these layers. The working principle is therefore from stack to stack (network to network) [32]. Because of its high depth, the network to network uses three Softmax classifiers. However, during the test phase, two auxiliary classifiers (Softmax) are left [33].

ResNet has a deep network structure that can efficiently overcome the backpropagation challenge and utilizes residual connections in its architecture. ResNet-50, as in GoogLeNet, is a type of a 50-layer network grouped in building blocks. In addition, this model was successful in classifying at the ImageNet competition in 2015. The network structure consists of a structure with increasing depth [34].

In this study, CNN architectures are used as a feature extractor. The reason for using CNN models only as feature extraction did not give the expected success as the same dataset was classified with the CNN models used in our previous study [35]. As the CNN models used in this study did not show good classification success, analysis results were not added. Important parameter information preferred in CNN models used in this study is as follows; the filter size of AlexNet was set to 3×3 pixels, the number of strides was two, and the max-pooling was preferred. The filter sizes of the GoogLeNet and ResNet-50 models were set to 7×7 pixels, the number of strides was two, and the selected pooling type was maximum. Also, the sizes of the input layers were adjusted to 224×224 pixels for GoogLeNet and ResNet-50. In addition, CNNs were trained without a transfer learning approach during the training.

3.3. Quadratic discriminant analysis (QDA) classifier

Discriminant analysis is a method generally used for classification and reduction of dimensionality. Linear Discriminant Analysis (LDA) is popular because it performs both classification and dimensionality reduction processes. As for QDA, one of the features that distinguish it from LDA is that it provides a non-linear analysis of data. QDA is a machine learning method that observes parameter values and predicts productively among labels, maximizing the common probability value. It is a type of discriminant analysis that predicts a separate covariance matrix for each observation class. The disadvantage of the QDA method is that it is not used as a dimensional reduction technique. The following equations are used in the QDA classification process [36]. In these equations, k represents each class and the total number of classes is expressed as $\sum k$. Here, k can be expressed as $k \in \{1, 2, \dots, k\}$. It is adjusted with π_k by the prevalence rate of observations specific to each class. The x in the equations represents observation, and the probability of x observation belonging to class k is determined by QDA. With QDA, new points are created by calculating by discriminant function or posterior probability count (δ_x). This calculation function is performed according to Eq. (1). Since the Covariance matrix is calculated for each class with QDA, the diagonal and upper right triangle of the matrix must be considered in this case. This upper region of the Covariance matrix is calculated according to Eq. (2). Since k such matrices need to be estimated, the calculation of covariance matrices is arranging according to Eq. (3). As a result, the number of effective QDA parameters is calculated according to Eq. (4). In these equations, the parameter μ_k is a class-specific mean vector. The number of QDA parameters is quadratic in p and p is an input parameter for class estimate [37–39].

$$\delta_x = -\frac{1}{2} \log \left| \sum k \right| - \frac{1}{2} (x - \mu_k)^T \sum_k (x - \mu_k) + \log \pi_k \quad (1)$$

$$\frac{p(p+1)}{2} \quad (2)$$

$$k \frac{p(p+1)}{2} \quad (3)$$

$$k - 1 + k_p + k \frac{p(p+1)}{2} \quad (4)$$

The reason for using the QDA classifier in this study was tested using the same dataset with different classifiers (kNN, LDA, SVM, etc.) in our previous study [35]. Since the QDA classifier gives more successful results, QDA was preferred in the direct classification process in this study.

3.4. Feature selection method

In machine learning, a feature selection method creates a subset of more discriminant features reducing computational cost

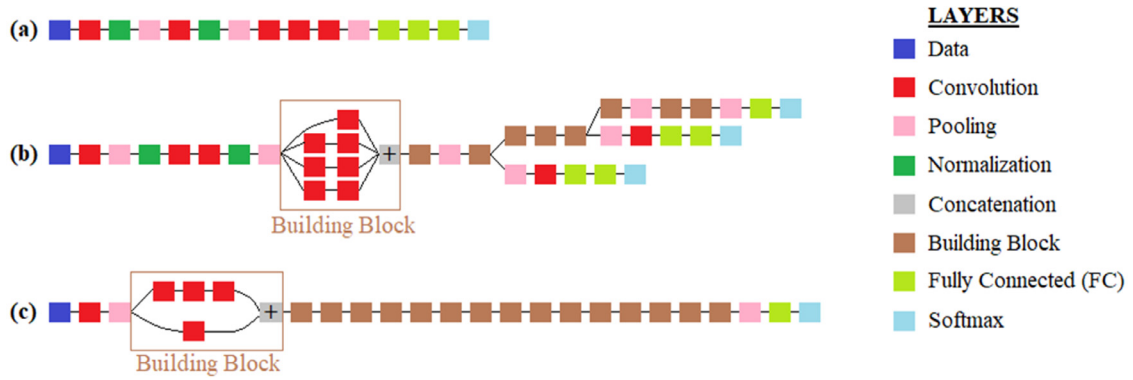


Fig. 2. The schematic representation of CNN models: (a) AlexNet, (b) GoogLeNet and (c) ResNet-50.

and increasing classification accuracy. In other words, feature selection is algorithms that reduce the dimensionality of data to improve the performance of machine learning methods. If this is exemplified, let us assume that the features obtained from a dataset are expressed by F and the dimensionality by S . When the feature selection method is applied to the data set, the number of new features obtained is expressed as F' and the dimensionality S' . The purpose of the feature selection method here is to provide the function $F' < F$ and $S' < S$. Therefore, feature selection is a widely used method to reduce dimensionality in data and to select features [40].

Two types of feature selection methods were used in this study. The first method is the Maximal Information Coefficient (MIC). This method examines the relationship between pairs of variables in large data sets. The mutual information measures the association between a pair of variables. It is a statistical-based method that measures the likelihood of coexistence by matching linear or nonlinear variables. The variables with maximal information are selected [41]. X and Y parameters are used as a variable in Eq. (5). Here the dimensions of the variables X and Y must be roughly the same. The values $X = 2$ and $Y = 2$ are generally accepted, but can be varied by $H(X)$. If the mapped variables are different in the dataset, normalization is performed by dividing the maximum value obtained in that set. Thus, the possible irregularity (entropy) among the variables is minimized [42]. This is shown in Eq. (6). The normalization of Eq. (6) is realized as in Eq. (7). The n_x and n_y parameters in Eq. (7) represent the normalization of X and Y .

$$H(X_b) = H(Y_b) = H(X_b, Y_b) \quad (5)$$

$$I(X; Y) = H(X) + H(Y) - H(X, Y) \quad (6)$$

$$\log(\min(n_x, n_y)) \quad (7)$$

Ridge regression is a statistical-based method that analyzes data based on multivariate cause-effect relationships. Ridge regression method is preferred because it gives lower values than the variance values from the Least Squares Method (LSM). Ridge Regression reduces the size of the regression coordinates. Thus, the effects of the problems are reduced. The Ridge regression formula is shown in Eq. (8). In this equation, z is a constant variable and takes a value between $0 < z < 1$. The unit matrix parameter is represented by I in Eq. (8) and the parameters X and Y represent the plane coordinates of each feature in the dataset [43,44].

$$(XX' + zI)^{-1} X'Y \quad (8)$$

In this study, the feature selection methods mentioned above were used in the proposed model. The reason for choosing MIC and Ridge Regression methods; Apart from these two feature

selection methods, Linear Regression, F Regression, and Recursive Feature Elimination (RFE) feature selection methods were also used [45]. The aim was to identify the first two feature selection methods that perform best in the five feature selection methods. Experimental analysis on the WBC dataset showed that MIC and Ridge regression methods are the two best performing feature selection algorithms. Source codes related to feature selection methods can be accessed from the reference section [46]. The MIC and Ridge regression methods were applied separately to the set of combined features. In this manner, the productive features were found. The aim here is to determine the common features, also referred to as intersecting features, selected by both these methods.

4. Proposed method

The proposed method consists of three basic steps. In the first step, the CNN models AlexNet, GoogLeNet, and ResNet-50 are used as feature extractors. After, the features obtained from the layer of CNN models were combined. In the second step, the feature selection methods MIC and Ridge Regression are employed. In the third step, the combination of feature selection methods has been realized. The intersecting features of MIC and Ridge Regression methods are classified by the QDA method. The general design of the proposed approach is showed in Fig. 3.

Other parameter selections of the CNN models used in this study are given in Table 1. In each iteration of mini-batch optimization, the model error is computed over sample subsets (batches) providing a good compromise between single sample usage and all sample usage [47]. If the computer's hardware resources are not sufficient, this may prevent the model from operating. Therefore, the mini-batch value was chosen as 16.

The followed training and evaluation algorithm steps are presented in Algorithm 1 to facilitate understanding of how the proposed model works.

5. Experimental evaluation

5.1. Experimental setting

We evaluate the proposed model through measures of sensitivity, specificity, F1-score and accuracy. The parameters used in these equations are; true positive (TP), true negative (TN), false positive (FP), false negative (FN) [48,49]. TP and TN represent correctly identified samples whereas FP and FN correspond to the incorrectly identified samples by the models.

$$\text{Sensitivity} = \frac{TP}{(TP + FN)} \quad (9)$$

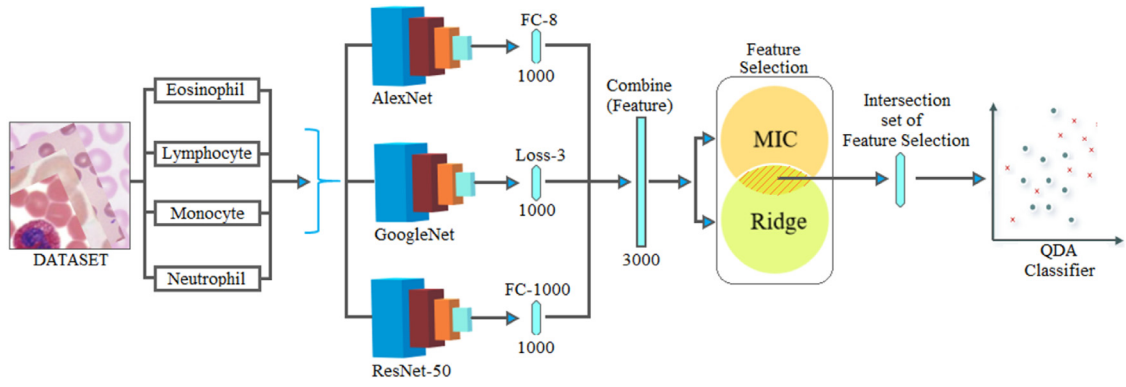


Fig. 3. Model design of the proposed method.

Algorithm 1: The training and evaluation algorithm of the proposed model

Input WBC images: training set and test set
(CNN Parameters)

- The CNN initial learning rate
- Iteration step
- The CNN maximum number of iteration
- The number of images covered in one iteration

Output Trained CNN model

Start

1. Initialize the CNN parameters
2. Light preprocessing the images with the necessary resolutions
3. Train the CNN and computing the weights
4. Deep feature extraction
5. Determining the most efficient deep features
6. Determining intersection set
7. Applying the selected features to QDA classifier
8. Performance evaluation

End

Table 1

Parameter values of CNN architectures.

Software used	CNN architecture	Image size	Optimization	Momentum	Decay	Beta	Mini batch	Learning rate
MATLAB	AlexNet	227 × 227	Sigmoid	0.9	1e−6	–	16	0.0001
	GoogLeNet	224 × 224	Gradient					
	ResNet-50	224 × 224	Descent					

$$\text{Specificity} = \frac{(TN)}{(TN + FP)} \quad (10)$$

$$\text{Precision} = \frac{(TP)}{(TP + FP)} \quad (11)$$

$$\text{F-score} = \frac{(2 \times TP)}{(2 \times TP + FP + FN)} \quad (12)$$

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FN) + (FP + TN)} \quad (13)$$

The experimental study was performed using MATLAB (R2019a) software installed on a 64-bit Windows 10 operating system. Python was used for MIC and Ridge regression methods [50]. The computer hardware was composed by an NVIDIA GeForce 2 GB graphics card, Intel © i5 - Core 2.5 GHz processor and 8 GB RAM. The experiments were divided into three categories.

5.2. Naive feature combination

In the first experiment, the dataset was trained using AlexNet, GoogLeNet, and ResNet-50 models, and the extracted feature maps were extracted. The FC-8 layer of the AlexNet model, the

loss-3 layer of the GoogLeNet model, and the FC-1000 layers of the ResNet-50 model were used in the feature extraction phase. Then, each feature set consisting of a totally 1000 extracted feature maps was classified with a QDA classifier. In the result of the first experiment, the ResNet-50 was determined as the most efficient. The success rate obtained in this model was 95.79%. In the next step, the features of these three CNN models were combined. As a result, we achieved a new feature set that consists of 3000 extracted feature maps. The combined feature set was classified with the QDA classifier. The classification accuracy was 95.79%. The analysis results of the first experiment are shown in Table 2.

5.3. Single feature selection method

In the second experiment, feature selection methods were used with the full feature set in the last stage of the first experiment. Features were selected from the full feature set by using MIC and Ridge feature selection methods. Each feature selection method selected 1000 features. While two feature selection methods select the productive features, they determine a random target column in the feature sets and normalize the features in the other columns. The purpose of normalizing feature

Table 2

The classification results of the dataset using the QDA method of CNN models. (Analysis results of the first experiment performed without using feature selection methods.)

CNN architecture	Class	Features	Precision	Sensitivity	F-Score	Accuracy (%)	Overall Acc. (%)
AlexNet	Eosinophil	1000	0.95	0.85	0.90	94.72	93.34
	Lymphocyte		0.96	0.99	0.97	98.76	
	Monocyte		0.93	1.00	0.96	98.22	
	Neutrophil		0.90	0.90	0.90	94.99	
GoogLeNet	Eosinophil	1000	0.92	0.85	0.88	93.99	91.24
	Lymphocyte		0.95	0.94	0.95	97.33	
	Monocyte		0.95	0.99	0.97	98.44	
	Neutrophil		0.83	0.87	0.85	92.72	
ResNet-50	Eosinophil	1000	0.96	0.93	0.94	97.06	95.79
	Lymphocyte		0.96	0.99	0.98	98.84	
	Monocyte		0.98	1.00	0.99	99.38	
	Neutrophil		0.94	0.92	0.93	96.31	
AlexNet & GoogLeNet & Resnet-50	Eosinophil	3000	0.92	0.93	0.92	96.25	95.79
	Lymphocyte		0.99	0.99	0.99	99.46	
	Monocyte		0.99	0.98	0.98	99.25	
	Neutrophil		0.94	0.93	0.93	96.63	

values is to facilitate the operation of feature selection algorithms. And each feature selection method performs the selection of efficient features by processing their algorithms according to the equations specified in Section 3.4. Here, the MIC and Ridge method scores between 0 and 1 for each feature column in the feature set. The feature column with the highest feature score is considered the most efficient feature [51,52]. QDA classifier was used for classification. The success rate obtained from the features extracted using the MIC method was 97.09%. The success rate obtained from selected features by Ridge method was 97.38%. The results are shown in Table 3. The feature selection methods used in the second experiment contributed to improving classification success.

5.4. Combining feature selection methods

In the third experiment, the feature selection methods MIC and Ridge Regression are combined. QDA was used for classification. For each method, the combined features were ranked according to the respective selection criteria. Then the first 500, 1000, 1500, and 2000 features of these rankings were used for each feature selection method. In the first step, an intersecting set of features was extracted between the top 500 features obtained from the MIC method and the top 500 features obtained from the Ridge method. The number of intersection features obtained from the 500 features of the two methods was 82. The success rate was 78.84% in the first step. In the second step, the common features, which are among the top 1000 features, were selected for both methods. The number of features that intersect in this step was 345. And the success rate of classification was 94.88% in the second step. In the third step, two feature selection methods were performed from among the best 1500 features. The number of intersection features was 755. The overall accuracy in classification performance was 97.95%. In the last step, common features between the best 2000 features of the two methods were extracted. The number of features that intersect in this step was 1343. The obtained overall accuracy rate was 95.90%. In the first three steps, the success of classification was increased. In the fourth step, the experiment was terminated because this performance was reduced. The analysis results of the third experiment are shown in Table 4.

As a result, the best achievement among all experiments was achieved in the third step of the third experiment. In this step, the accuracy rates of the four classes separately were obtained as 98.19% in Eosinophil, 99.78% in Lymphocyte, 99.81% in Monocyte, and 98.11% in Neutrophil. Classification performance measures (the sensitivity, precision, etc.) were taken into consideration [53–55], the overall accuracy rate was 97.95%. The confusion

matrix of the third step and the ROC curve graph are shown in Fig. 4. In addition, intersecting features were extracted by the feature selection methods. This shows that more intersecting features were selected. The performance graph of all experiments is shown in Fig. 5.

6. Discussion

WBC is a stabilizing factor in the immune system of the human body. WBC counting outside the normal range is considered a symptom of various diseases. If the related diseases are not diagnosed in time, they can cause permanent damage to the human body, even death. Early and accurate detection of WBC is associated with the fast and accurate results of image processing techniques. Conventional methods and techniques (Gabor Wavelets, Random Forest, LBP, Principal Component Analysis, etc.) can be used to improve the efficiency of CNN models. As a result, numerous studies have been conducted based on machine learning and deep learning models in the classification of WBC types. Experimental results of our study, the combination of feature selection methods yielded a higher accuracy (97.95%) compared to simple feature combination (95.79%). To obtain better performance from CNN models in accordance with the experimental analysis performed in this study, it is necessary to pass the images through a preprocessing process (attention modules, segmentation, etc.). In this context, preprocessing steps in the literature obtained using the same and similar dataset support this situation. The dataset used in this study was also used by other researchers and they were classified with different models/methods. The results of other studies using the same or similar datasets are shown in Table 5.

G. Liang et al. [56] used at the WBC dataset classification by combining the RNN model with the CNN model. They used CNN models, such as Inception V3, ResNet-50, and Xception. G. Liang et al. [56] created image patches from the dataset they used. They obtained image patches using the rotation matrix for each image. The aim was to improve the classification performance with these patches. As a result, the success rate of CNN & RNN combination was 90.79%. In our study, CNN models were trained from scratch. G. Liang et al. [56] used pre-trained CNN models contributed to the success of classification in their study. If we preferred pre-trained CNN models in the study in this article, we could increase the accuracy of classification. The better accuracy obtained with our approach can be attributed to the combination of feature selection methods.

D. Bani-Hani et al. [57] used the genetic algorithm, which is the optimization method, together with the CNN model. They

Table 3

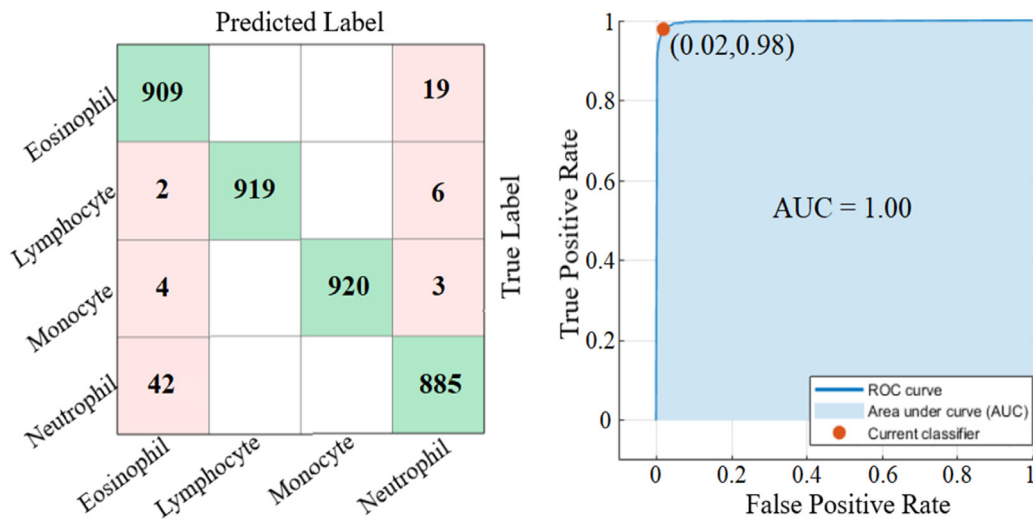
Classification of the combined feature set (total 3000 features) using feature selection methods separately. (Analysis results of the second experiment.)

CNN architectures	Class	Method & Features	Precision	Sensitivity	F-Score	Accuracy (%)	Overall Acc. (%)
AlexNet & GoogLeNet & Resnet-50	Eosinophil	MIC & 1000	0.98	0.92	0.95	97.30	97.09
	Lymphocyte		0.99	1.00	1.00	99.84	
	Monocyte		0.98	1.00	0.99	99.60	
	Neutrophil		0.93	0.97	0.95	97.44	
AlexNet & GoogLeNet & Resnet-50	Eosinophil	Ridge & 1000	0.99	0.92	0.95	97.63	97.38
	Lymphocyte		0.99	1.00	1.00	99.81	
	Monocyte		0.98	1.00	0.99	99.62	
	Neutrophil		0.93	0.98	0.95	97.71	

Table 4

Classification of feature sets extracted by MIC and Ridge methods by finding intersecting features. (Analysis results of the third experiment.)

Method/Top features	Class	Intersecting features	Precision	Sensitivity	F-Score	Accuracy (%)	Overall Acc. (%)
MIC/500 & Ridge/500	Eosinophil	82	0.70	0.73	0.71	85.98	78.84
	Lymphocyte		0.83	0.84	0.83	91.61	
	Monocyte		0.91	0.82	0.87	92.96	
	Neutrophil		0.71	0.76	0.73	87.11	
MIC/1000 & Ridge/1000	Eosinophil	345	0.92	0.90	0.91	95.44	94.88
	Lymphocyte		0.99	0.99	0.99	99.43	
	Monocyte		0.98	1.00	0.99	99.54	
	Neutrophil		0.90	0.91	0.91	95.34	
MIC/1500 & Ridge/1500	Eosinophil	755	0.98	0.95	0.96	98.19	97.95
	Lymphocyte		0.99	1.00	1.00	99.78	
	Monocyte		0.99	1.00	1.00	99.81	
	Neutrophil		0.95	0.97	0.96	98.11	
MIC/2000 & Ridge/2000	Eosinophil	1343	0.98	0.89	0.93	96.47	95.90
	Lymphocyte		0.99	1.00	0.99	99.70	
	Monocyte		0.95	1.00	0.98	98.84	
	Neutrophil		0.91	0.96	0.93	96.79	

**Fig. 4.** The confusion matrix and ROC curve of the best analysis result of the proposed approach.**Table 5**

Comparison of this study with other studies using the same and similar dataset.

Study	Year	Dataset	Model/Method	Acc. (%)
G. Liang et al. [56]	2018	Same	CNN & Recurrent Neural Network (RNN)	90.79
D. Bani-Hani et al. [57]	2018	Same	CNN & Genetic Algorithm (GA)	91.00
R.B. Hegde et al. [58]	2019	Similar	CNN & LBP	99.00
M. Sharma [10]	2019	Similar	CNN & Data Augmentation approach	87.00
L. Wang [11]	2018	Same	PatternNet-fused Ensemble of CNN	99.90
M. Habibzadeh [9]	2013	Similar	Histogram features, kernel principal component, CNN and SVM	85.00
Proposed approach	2020	Same	CNN & Feature selection & Intersection of features in sets	97.95

set the data input size of the CNN model that they designed in their study to 32×32 . Although this situation accelerated the training process of the CNN model, it was thought that features may have a negative effect. The CNN architecture consisted of six

convolutional, two pooling, and intermediate layers. Due to the insufficiency of the hardware units used in their studies, the GA population number was not enough. As a result, their success rate was limited to 91%. The input values of the CNN models used

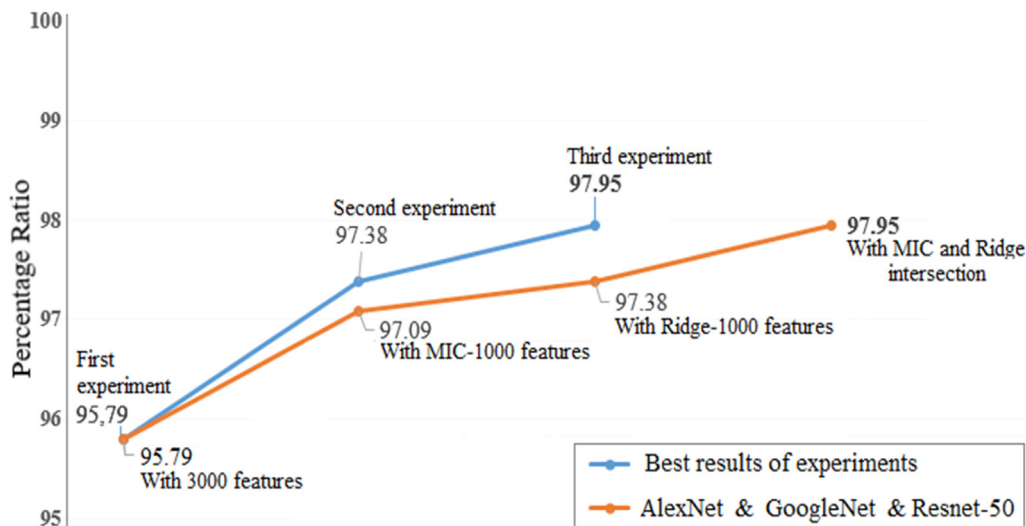


Fig. 5. Performance graph of the proposed approach on the experiments performed in this study.

in the study of this article are 224×224 and 227×227 . These values are extracted more productive features than D. Bani-Hani et al. [57] (32×32) data entry. Therefore, the classification success in their study is lower.

R. B. Hegde et al. [58] classified five types of WBC images with the CNN model in their study. They used the AlexNet model as the CNN model. After passing the dataset through the preprocessing steps with the LBP technique, they obtained feature sets from the images using the AlexNet model. They obtained the feature sets from the FC6, FC7, and FC8 layers of the AlexNet model. They achieved the most successful result in the classification performance with 1000 feature sets from the FC8 layer. The average accuracy success in their study was 99%. The dataset they used in this study was not the same as our dataset, and they used a five-class dataset by including “Basophil” images in their study. They found that they significantly improved the detection performance on the images with the LBP technique. However, the number of images in the dataset they use in their study is very low compared to their types. The number of images in the dataset that we use in our study is sufficient to perform experimental steps. This may question the accuracy of the analysis results. In addition, R. B. Hegde et al. [58] stated that they set the test rate of the dataset to 20% in their study. But the values in the confusion matrix they show in their study show that they use all of the data in the dataset. We think that this contradiction in their study may have been cross-validated in the classification process. As a result, with the proposed approach of our study, it is both innovative and has reliable results since the number of images in its dataset is sufficient.

7. Conclusion

In this study, we evaluate the combination of feature selection methods for WBC classification with CNNs. With the proposed approach, the features obtained from 1000-feature layers of AlexNet, GoogLeNet, ResNet-50 architectures were combined. MIC and Ridge feature selection methods were used to select features in the combined feature set. It was seen that the intersecting features from the feature sets obtained from the MIC and Ridge methods contributed to the classification results. Here, it was proved that the intersecting features of the two feature selection methods are more successful than a single feature selection method. Experimental results show that the overall accuracy rate obtained with 755 intersecting features as a result of the

classification made by the QDA method was 97.95%. The accuracy achievements of WBC types on a class basis; It was 98.19% in eosinophil, 99.78% in lymphocyte, 99.81% in monocyte, and 98.11% in neutrophil.

Our future study is planned to be developed by applying to preprocess steps (segmentation, reconstructing the image, attention modules, etc.) to the proposed approach. Thus, the proposed approach can be used to design high-performance computer-assisted systems for other future medical imaging tasks.

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Ethical approval

This article does not contain any data, or other information from studies or experimentation, with the involvement of human or animal subjects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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